PROGRAMMING 742

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ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

☐ Use Times New Roman font, size 12, with 1.5 line spacing throughout the document.

☐ Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

☐ Table of Contents

☐ Introduction

☐ Main Body (with relevant subheadings)

☐ Conclusion

☐ References

☐ Submit the assignment in PDF format on Moodle.

☐ Use the specified cover page provided.

☐ Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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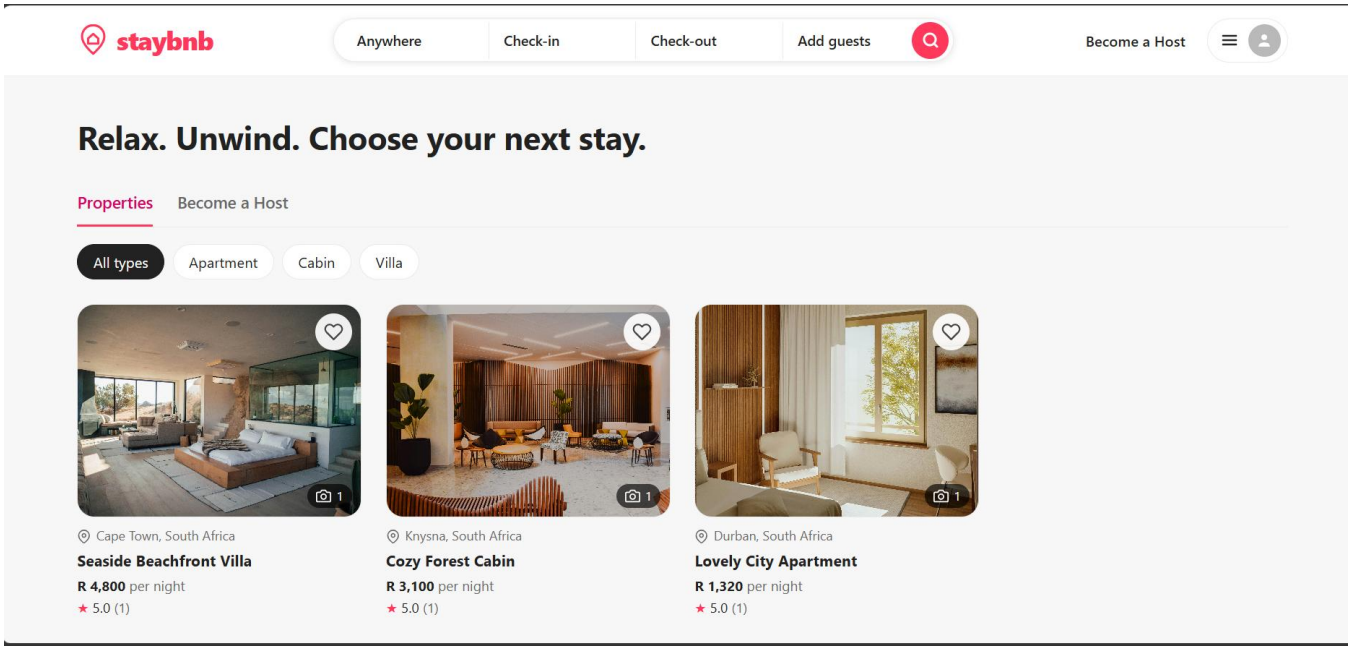
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1. Overview

Staybnb is a managed property rental website built with ASP.NET Core MVC, Entity Framework Core and SQL Server. It covers the full user lifecycle: a person registers as a Guest, explores and books stays, applies to become a Host, and Administrators review applications and keep the marketplace safe. The site follows the layout of the screenshots in the brief: an Airbnb-style gallery for guests and sidebar dashboards for hosts and admins.

HomePage:



2. Technology

ASP.NET Core MVC (.NET 10) with Razor views and Bootstrap 5

Entity Framework Core with SQL Server LocalDB, using code-first migrations

ASP.NET Core Identity with roles (SuperAdmin, Admin, Host, Guest)

Role protection with [Authorize(Roles = "...")] on every dashboard and workflow controller

3. How to run the project

Open Staybnb.sln in Visual Studio and press the green Run button.

No manual database steps are needed. On start-up the app applies the EF Core migrations, creates the four roles, the SuperAdmin, 12 amenities and demo data (RoleSeeder and DataSeeder).

If the database ever needs a reset: Package Manager Console, run Drop-Database, then run the app again.

Ready-made logins

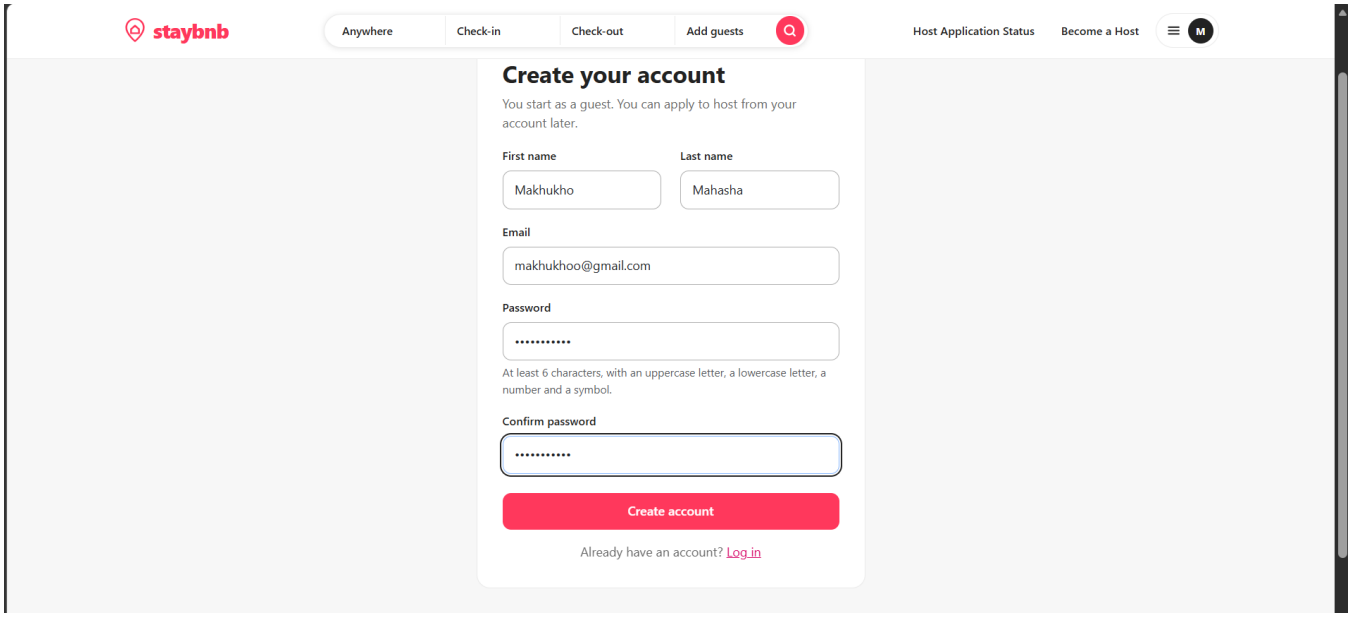
Role	Email	Password
SuperAdmin	superadmin@staybnb.com	Admin@12345
Host (demo)	host@staybnb.com	Host@12345
Guest (demo)	guest@staybnb.com	Guest@12345

4. Requirements and where they are met

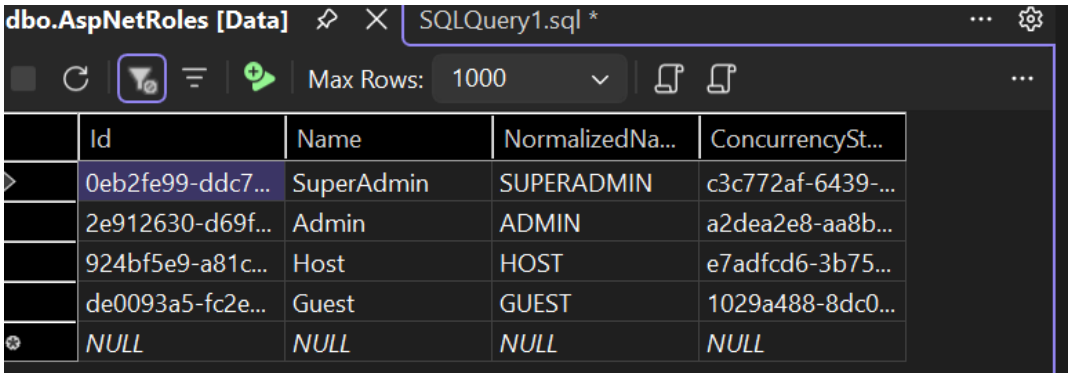
4.1 Authentication and role logic

Register and login are built on ASP.NET Core Identity. Every new account becomes a Guest. A RoleSeeder creates the four roles and the SuperAdmin on start-up. When an Admin approves a HostApplication the system adds the Host role and removes the Guest role using UserManager, then refreshes the security stamp so the user signs in again with the new role. Only the SuperAdmin can promote a Guest to Admin.

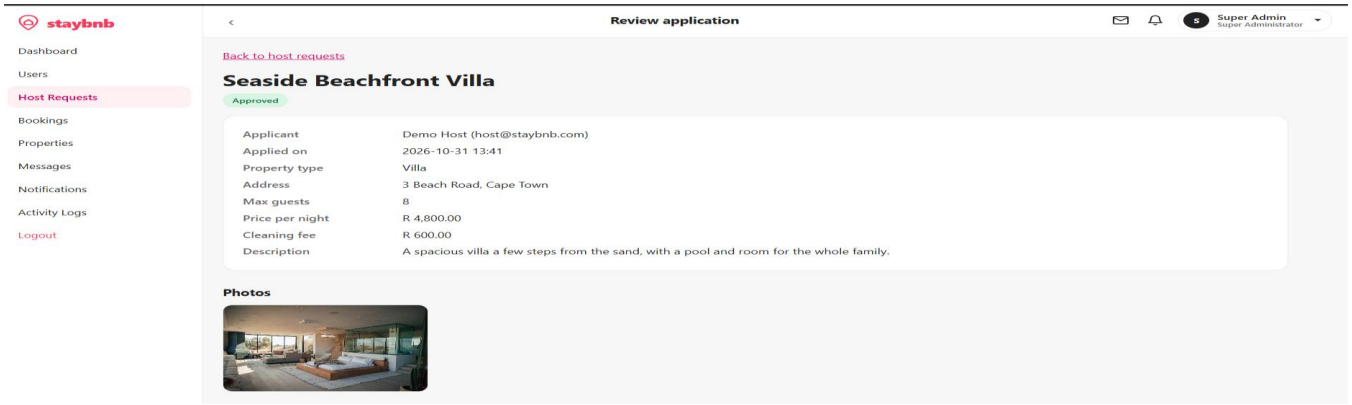
Sign-up Page



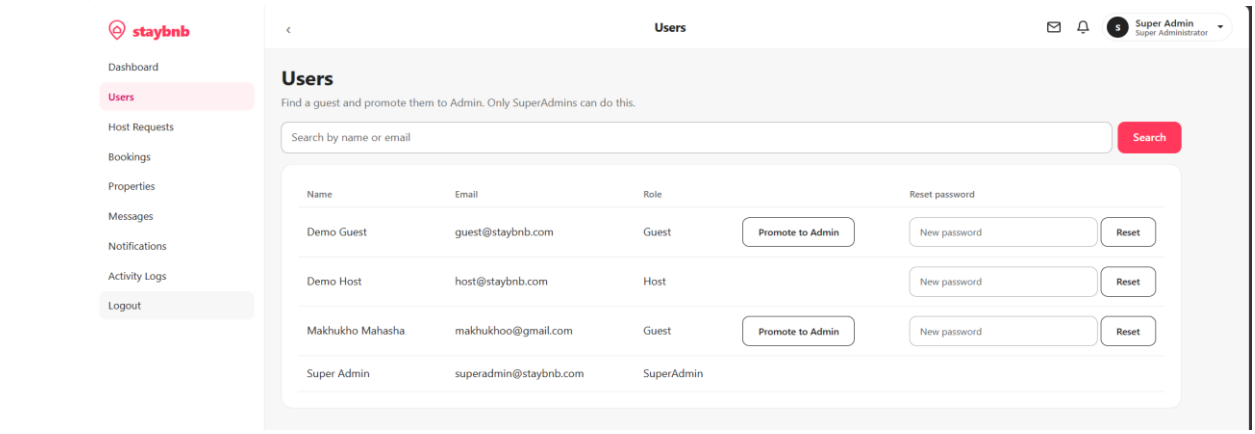
The four roles stored in the database (AspNetRoles table)



Admin reviewing a host application with the approve button



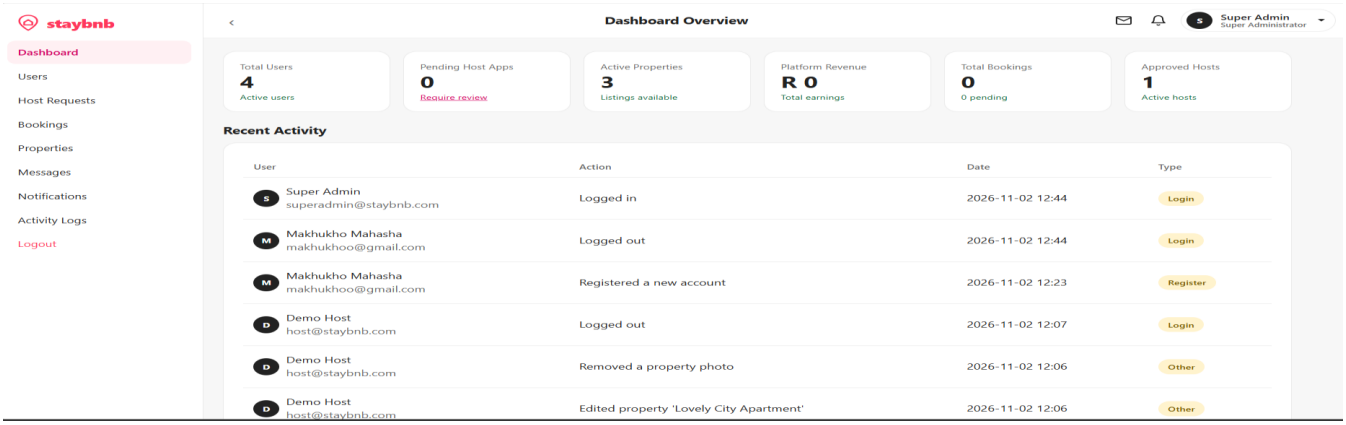
Users page showing each person’s single role (Guest, Host, Admin, SuperAdmin)



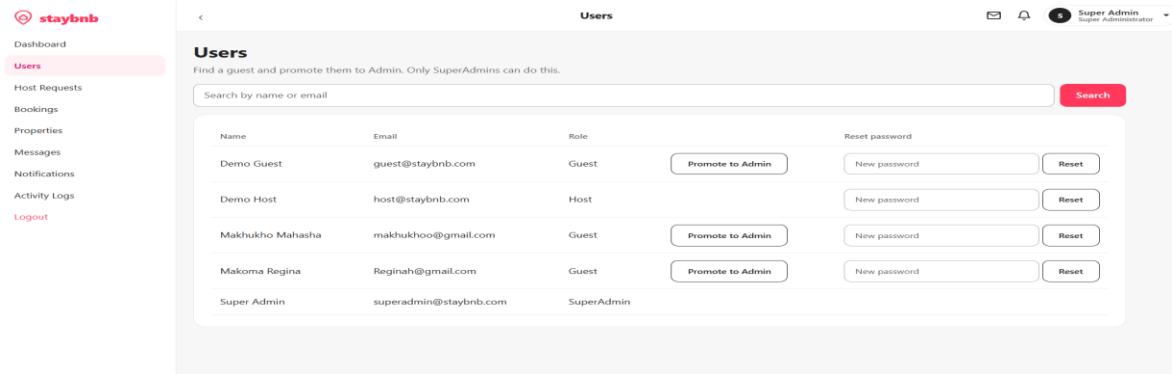
4.2 Admin dashboard

The Admin dashboard shows statistics, recent activity, recent host applications and recent bookings. Admins approve or reject applications, see all bookings and properties, and read the Activity Log. The SuperAdmin also gets a Users page to find Guests, promote them to Admin and reset passwords.

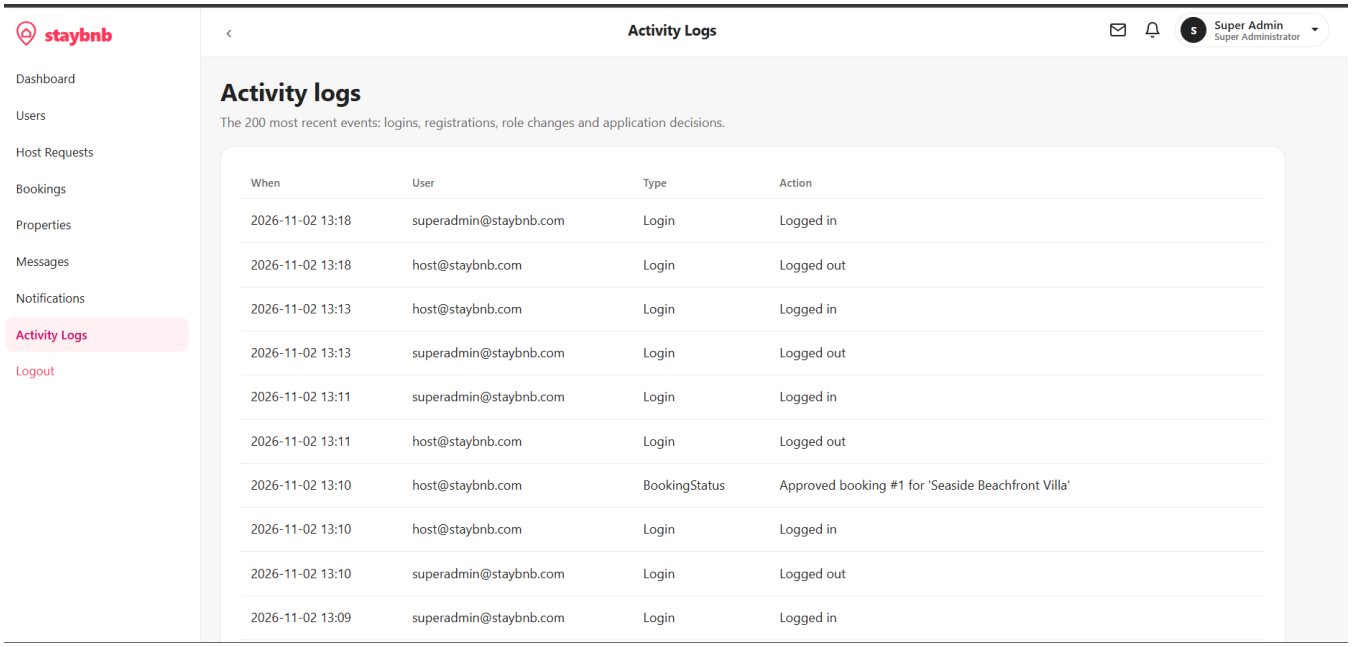
Admin Dashboard Overview with the six statistics cards and recent tables



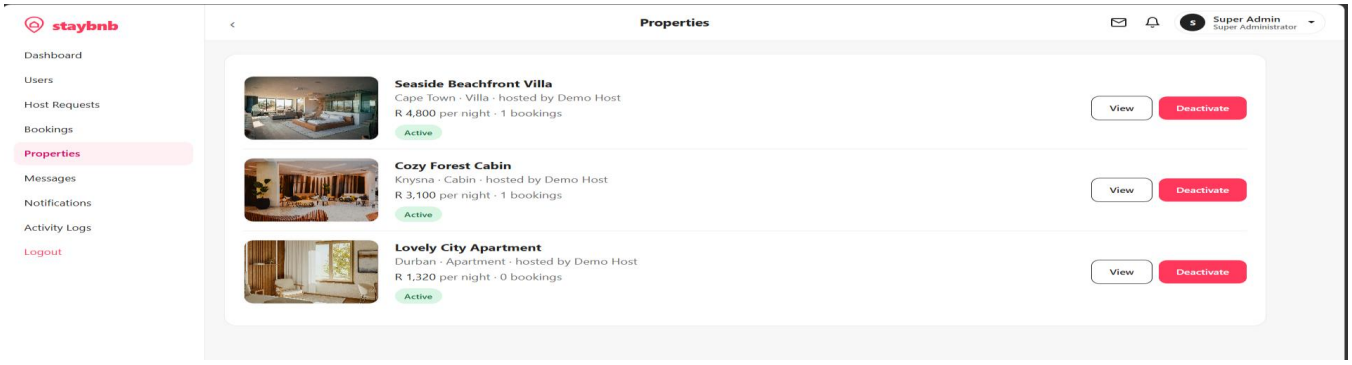
Users page with the promote to admin Button



Activity logs showing logins, role changes and booking status changes



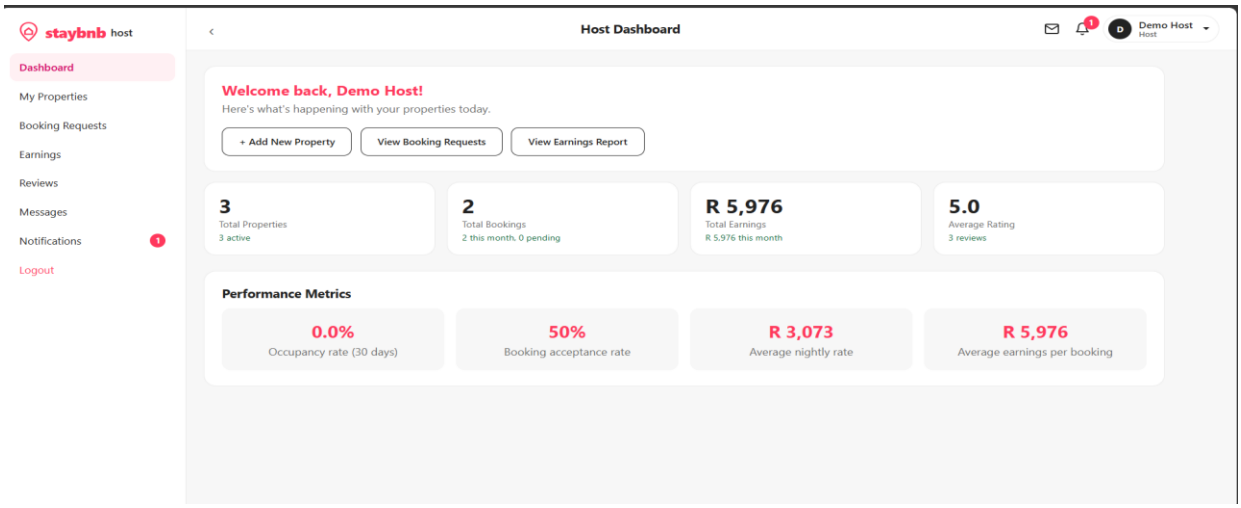
Admin Bookings or properties page



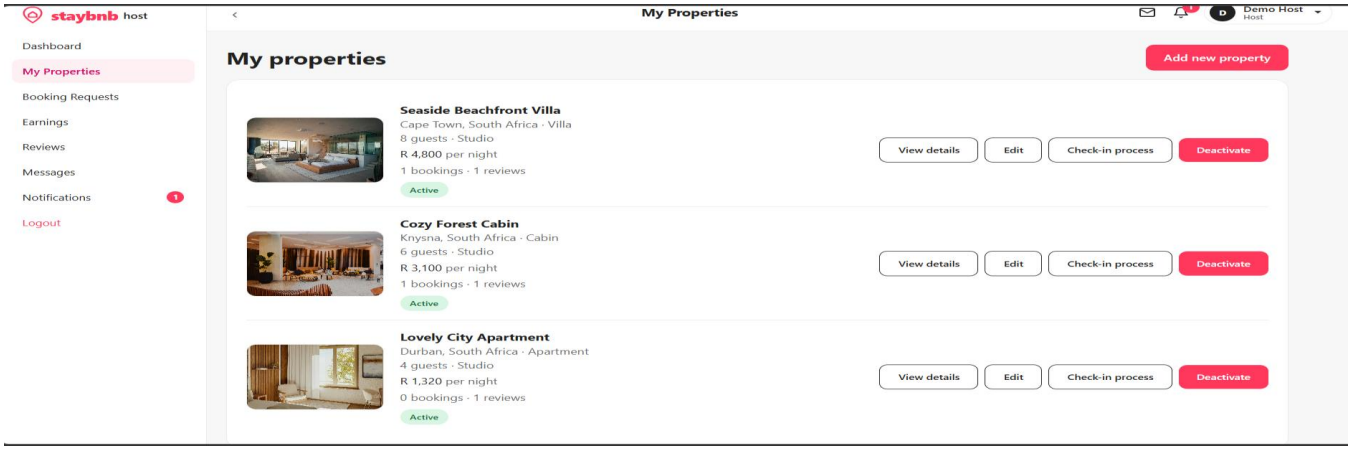
4.3 Host dashboard

Once approved, the user lands on a Host dashboard with a sidebar. Hosts manage their properties and switch IsActive on or off, approve or reject booking requests, define the CheckInProcess (house rules and the documents guests must upload) and verify or reject the documents guests upload.

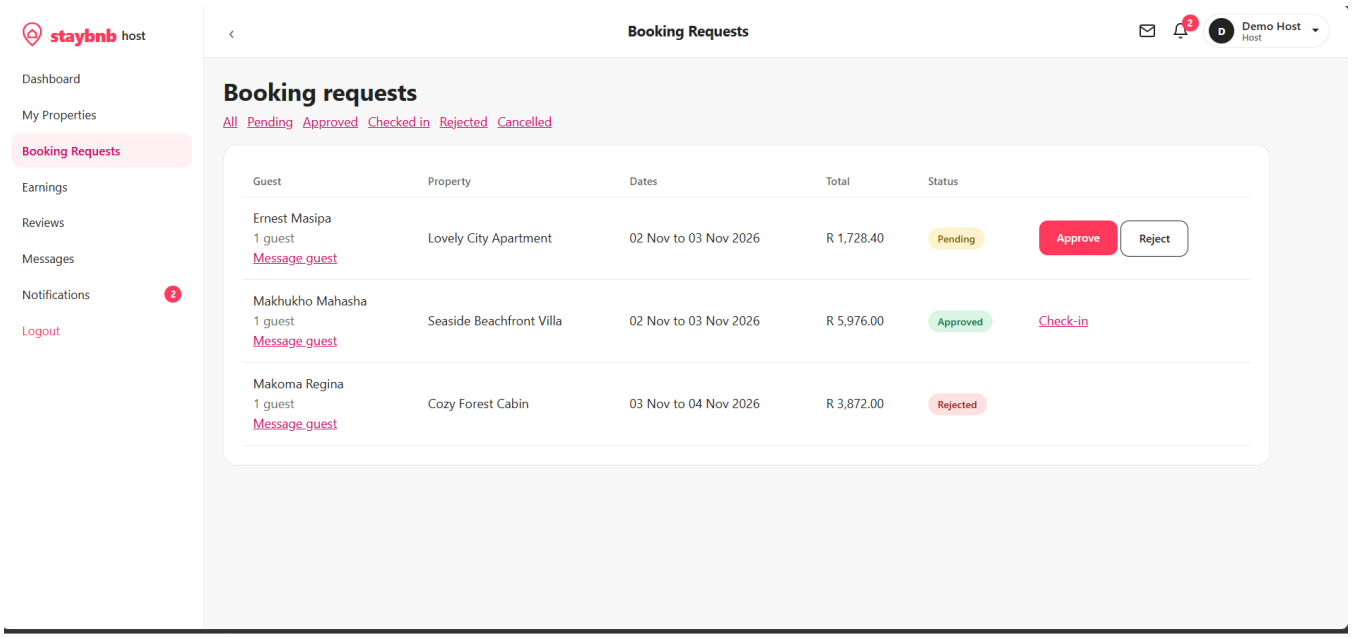
Host dashboard with welcome card, statistic cards and performance metrics



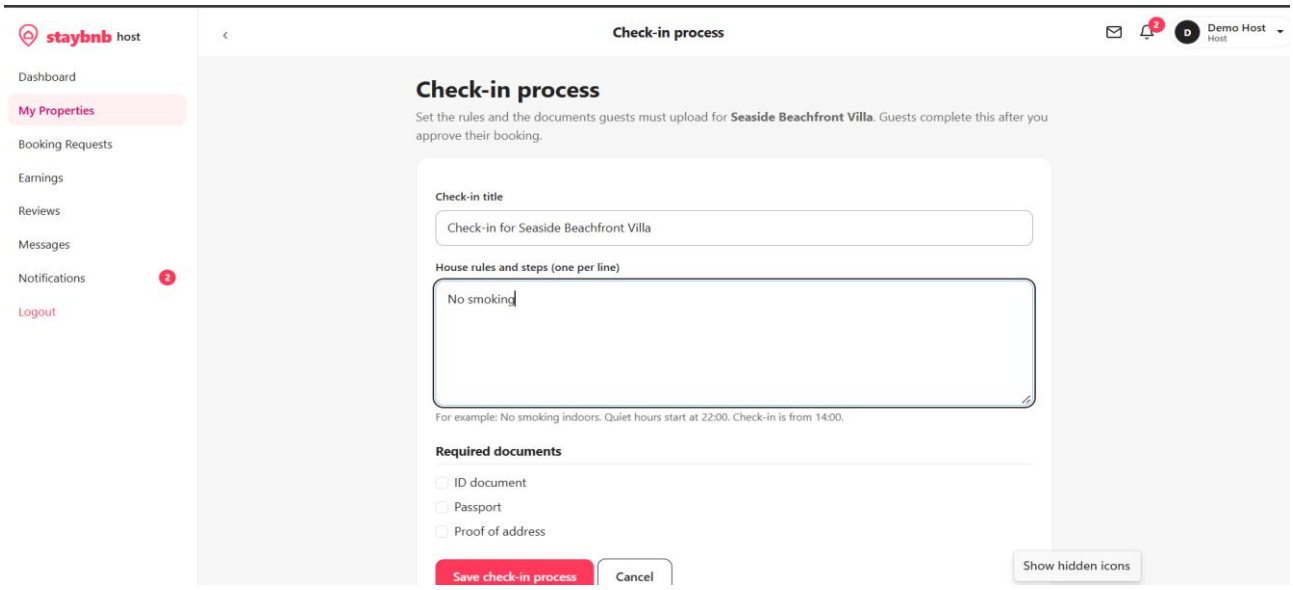
My properties list with Active and InActive status and the Deactivate or Activate Button.



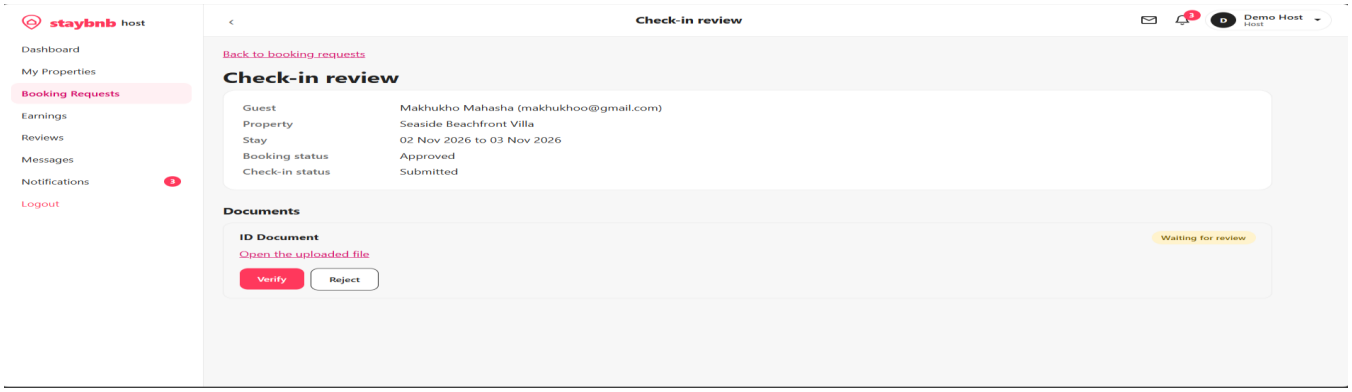
Booking Requests with Approve and Reject buttons on a pending booking



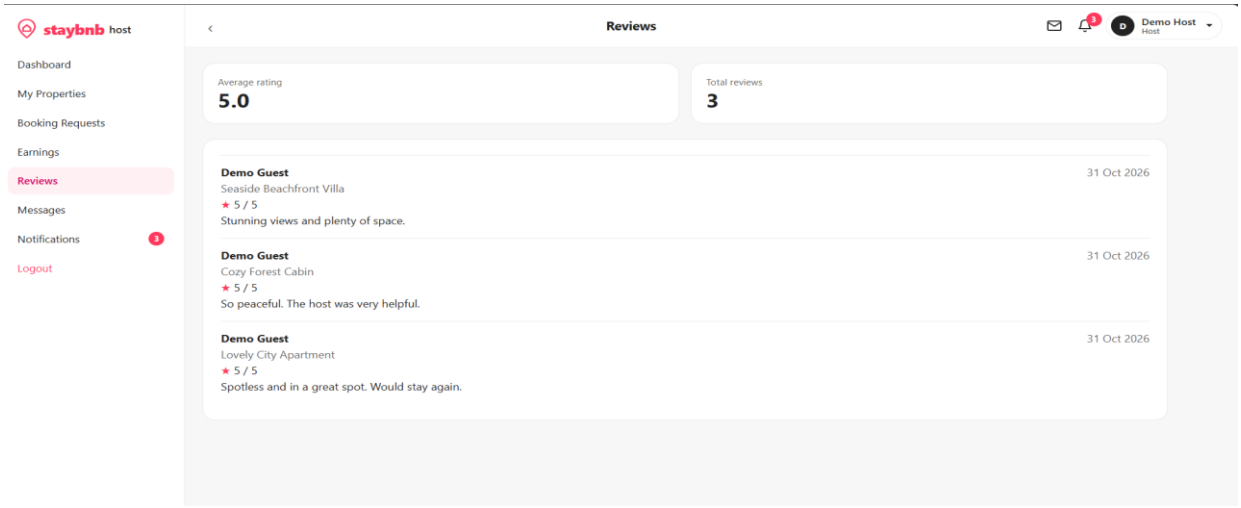
Check-in process setup: rules and required documents



Host verifying or rejecting a guest's uploaded document



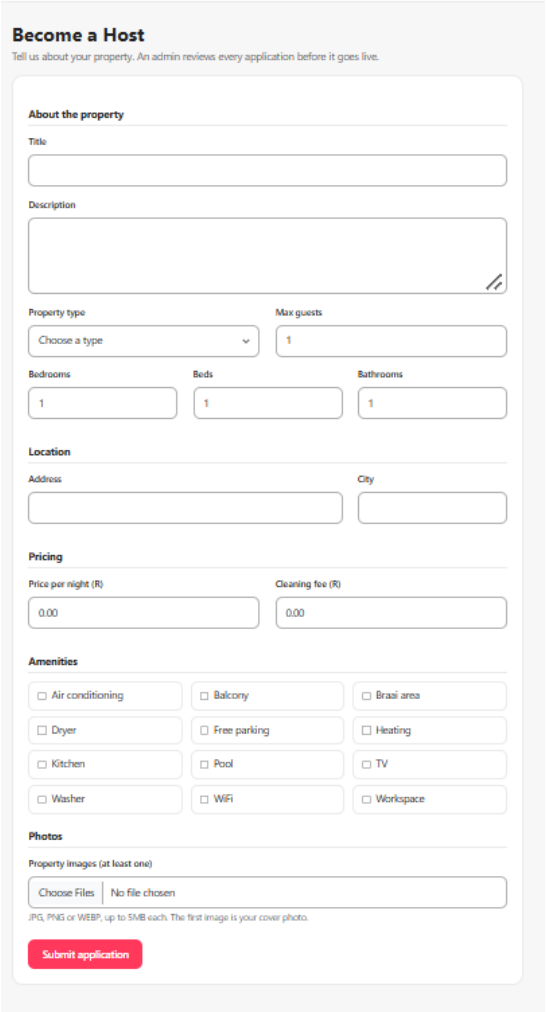
Review page



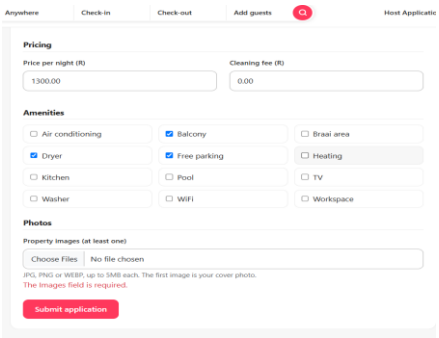
4.4 Application process

The Become a Host form collects the property details (title, description, price, address, city, type, rooms, amenities) and photos. The property, its amenities, its images and the HostApplication are saved together in one transaction. An application cannot be submitted without at least one image (JPG, PNG or WEBP, up to 5MB).

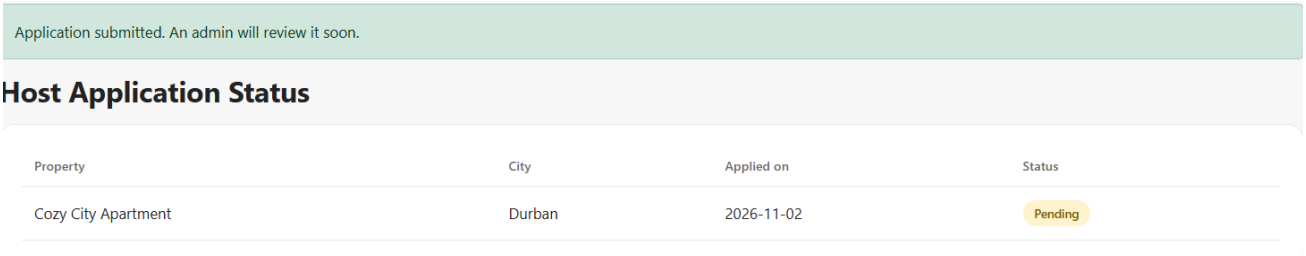
Become a Host form



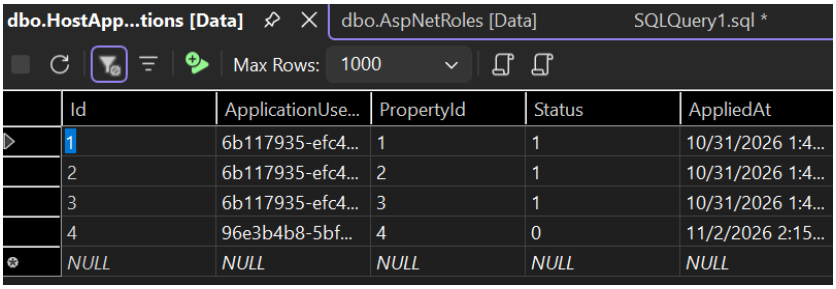
Validation message “Please upload at least one image” when no photo is chosen.



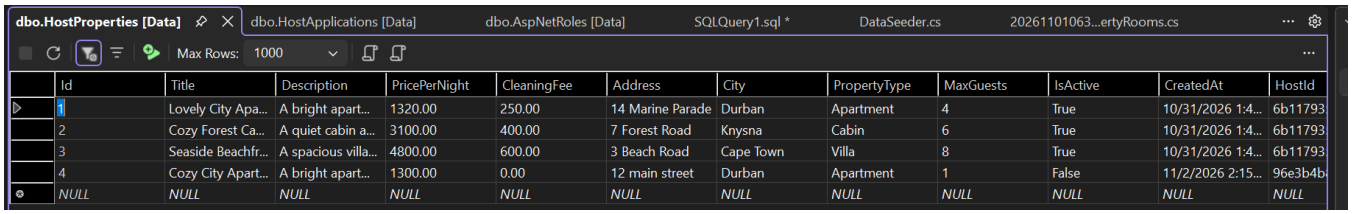
Host Application Status showing the Pending badge



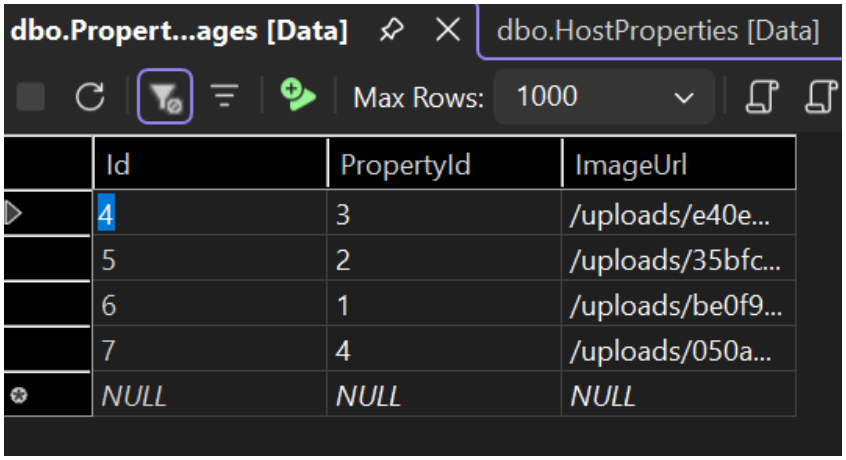
Database rows created by one application: HostApplications, HostProperties and PropertyImages



HostProperties



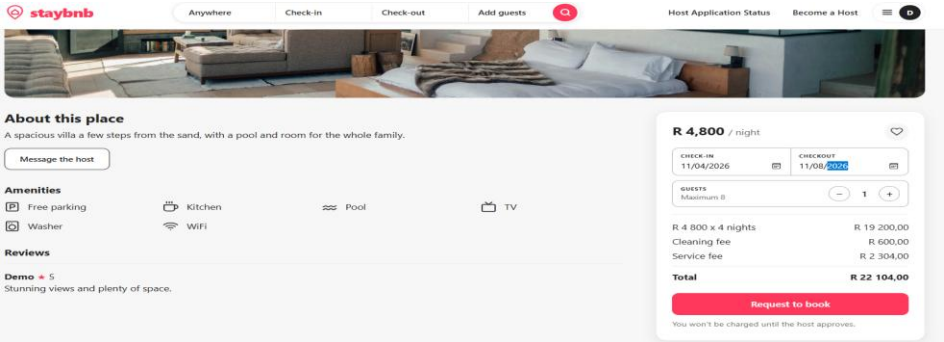
PropertyImages



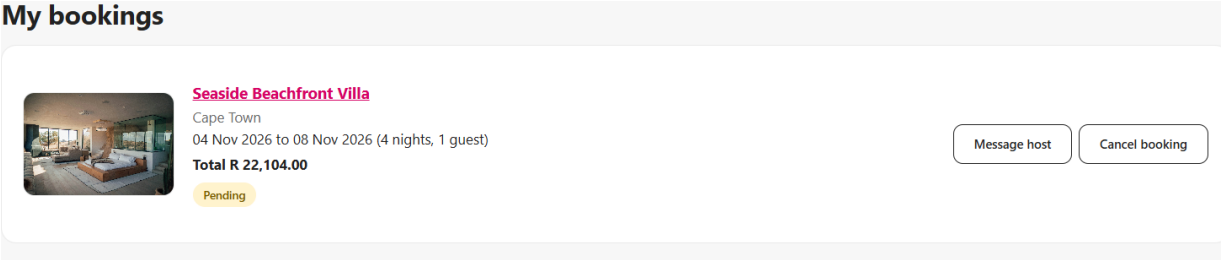
4.5 Booking and check-in

Guests choose dates and the number of guests. The server calculates TotalPrice = (PricePerNight x Nights) + CleaningFee + ServiceFee, where the service fee is 12 percent of the nightly total, and it blocks overlapping dates. The status flow is Pending, Approved, CheckedIn. After approval the guest completes a three-step check-in (house rules, document upload, submit) and the host verifies the documents.

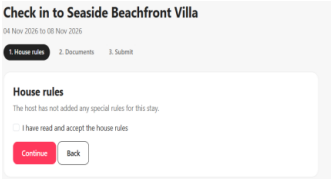
Property page with the booking card and the price breakdown (nights, cleaning fee, service fee, total)

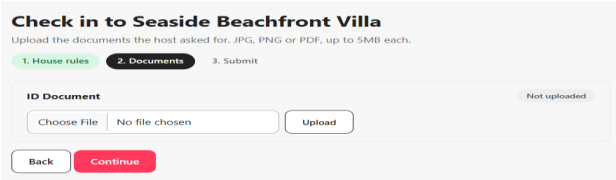


My bookings with a pending booking and its total

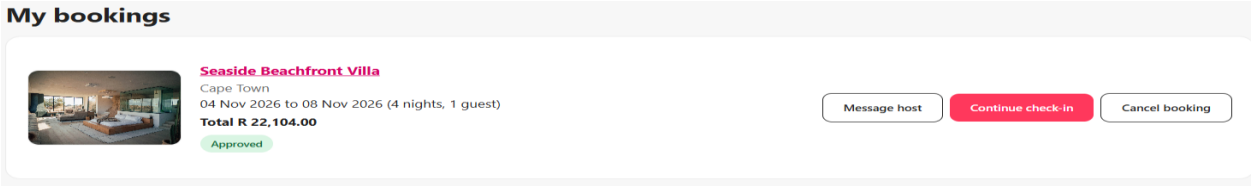


Check in step 1 (house rules) and step 2 (document upload)

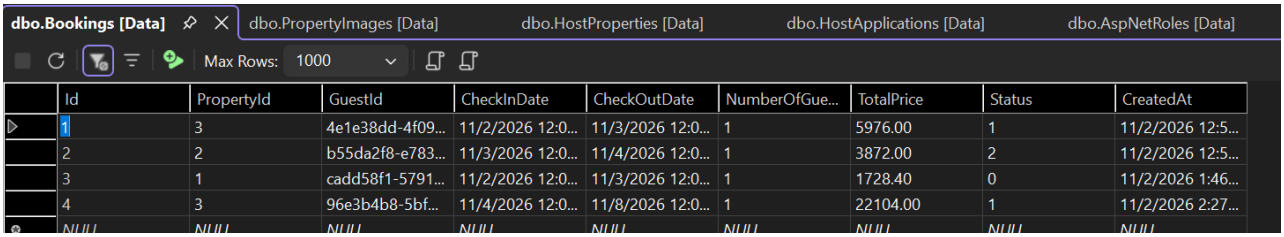




Booking showing the checked in status after the host verified the documents



Booking table

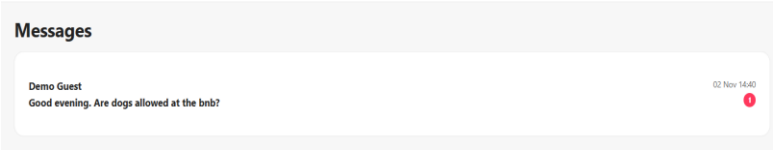


Id	PropertyId	GuestId	CheckInDate	CheckOutDate	NumberOfGue...	TotalPrice	Status	CreatedAt
1	3	4e1e38dd-4f09...	11/2/2026 12:0...	11/3/2026 12:0...	1	5976.00	1	11/2/2026 12:5...
2	2	b55da2f8-e783...	11/3/2026 12:0...	11/4/2026 12:0...	1	3872.00	2	11/2/2026 12:5...
3	1	cadd58f1-5791...	11/2/2026 12:0...	11/3/2026 12:0...	1	1728.40	0	11/2/2026 1:46...
4	3	96e3b4b8-5bf...	11/4/2026 12:0...	11/8/2026 12:0...	1	22104.00	1	11/2/2026 2:27...
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

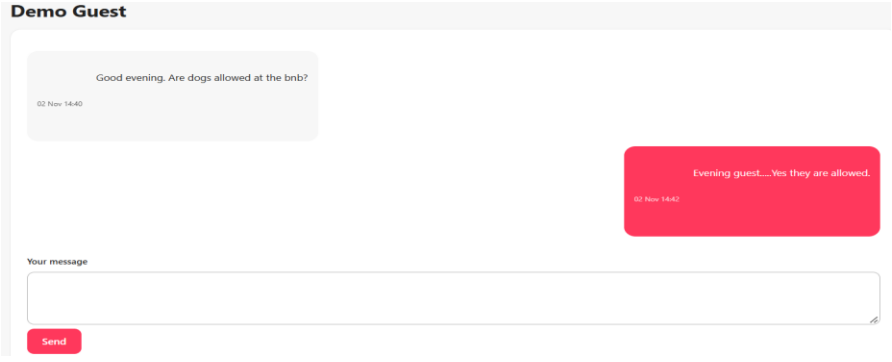
4.6 Messaging and notifications

Messaging is database-backed. Each user has an inbox of conversations and can read and reply; opening a conversation marks incoming messages as read. A bell icon with an unread count opens the notification centre. Opening a notification marks it as read in the database.

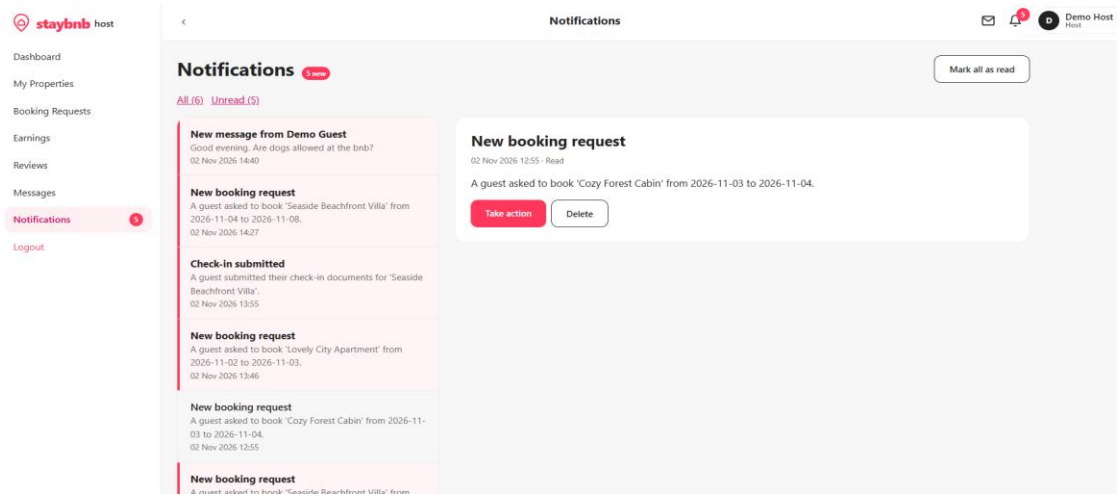
Inbox showing a conversation with an unread count



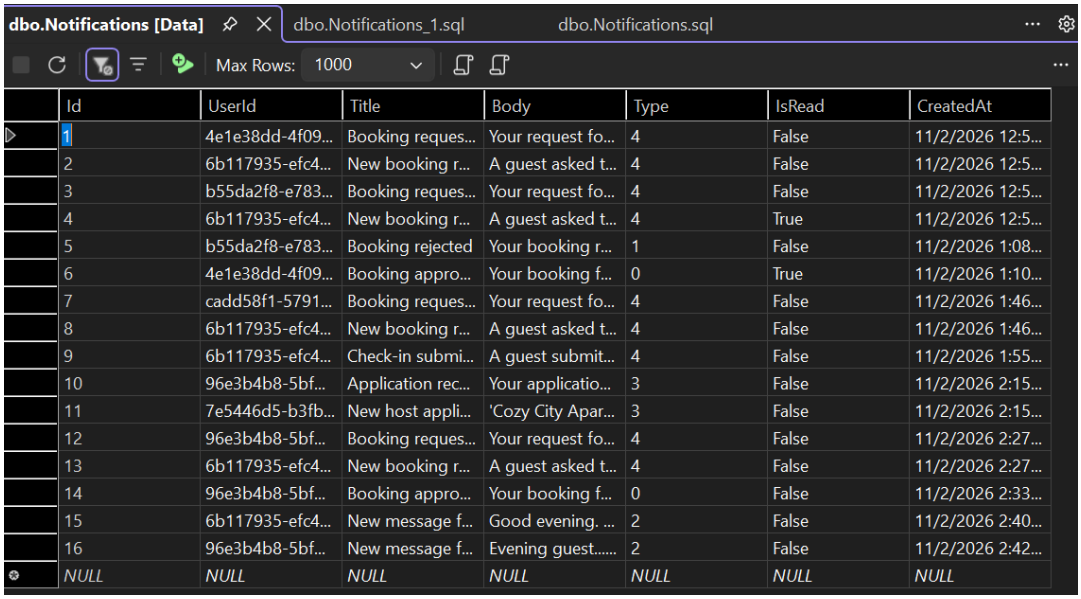
A conversation with a reply



Notifications page with an unread item and the opened item on the right



Notifications or messages table showing IsRead False and then True after opening.

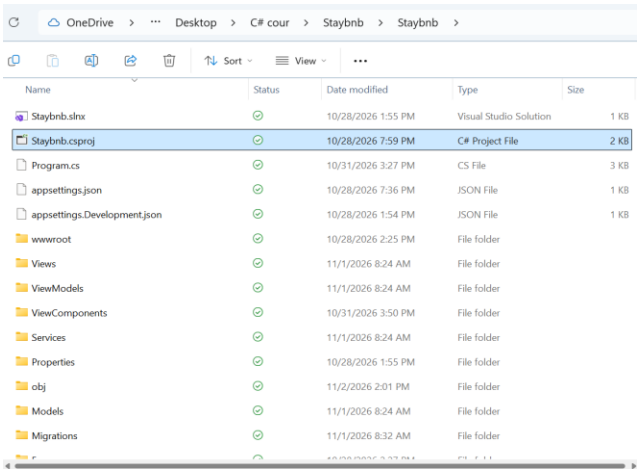


Id	UserId	Title	Body	Type	IsRead	CreatedAt
1	4e1e38dd-4f09...	Booking reques...	Your request fo...	4	False	11/2/2026 12:5...
2	6b117935-efc4...	New booking r...	A guest asked t...	4	False	11/2/2026 12:5...
3	b55da2f8-e783...	Booking reques...	Your request fo...	4	False	11/2/2026 12:5...
4	6b117935-efc4...	New booking r...	A guest asked t...	4	True	11/2/2026 12:5...
5	b55da2f8-e783...	Booking rejected	Your booking r...	1	False	11/2/2026 1:08...
6	4e1e38dd-4f09...	Booking appro...	Your booking f...	0	True	11/2/2026 1:10...
7	cadd58f1-5791...	Booking reques...	Your request fo...	4	False	11/2/2026 1:46...
8	6b117935-efc4...	New booking r...	A guest asked t...	4	False	11/2/2026 1:46...
9	6b117935-efc4...	Check-in submi...	A guest submit...	4	False	11/2/2026 1:55...
10	96e3b4b8-5bf...	Application rec...	Your applicatio...	3	False	11/2/2026 2:15...
11	7e5446d5-b3fb...	New host appli...	'Cozy City Apar...	3	False	11/2/2026 2:15...
12	96e3b4b8-5bf...	Booking reques...	Your request fo...	4	False	11/2/2026 2:27...
13	6b117935-efc4...	New booking r...	A guest asked t...	4	False	11/2/2026 2:27...
14	96e3b4b8-5bf...	Booking appro...	Your booking f...	0	False	11/2/2026 2:33...
15	6b117935-efc4...	New message f...	Good evening. ...	2	False	11/2/2026 2:40...
16	96e3b4b8-5bf...	New message f...	Evening guest.....	2	False	11/2/2026 2:42...
NULL	NULL	NULL	NULL	NULL	NULL	NULL

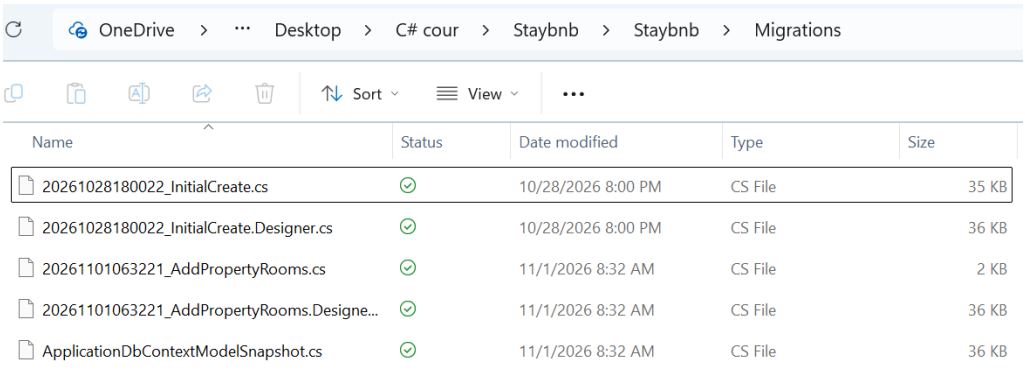
4.7 Data and architecture

The EF Core migrations create the whole schema. ActivityLog entries are written for logins, registrations, role changes, applications and booking status changes. The code is organised into Controllers, Models, ViewModels, Services, Data (DbContext and seeders), ViewComponents and Views, with small services (ActivityLogger, NotificationService, BookingPricing).

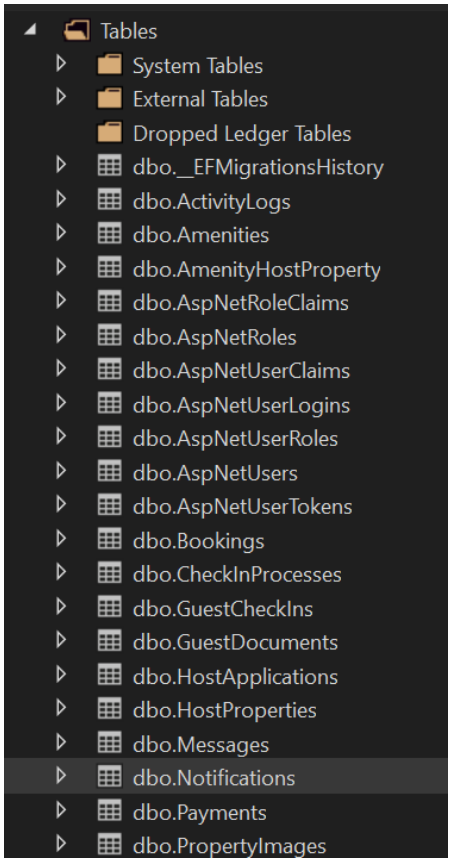
Explorer showing the project folders



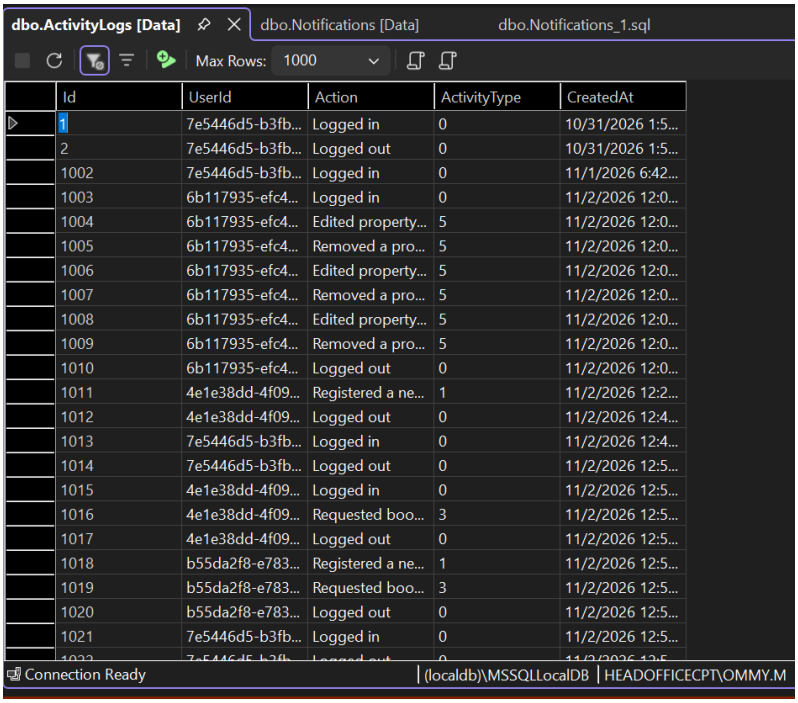
Migrations Folder



All database tables



ActivityLogs table with rows



4.8 Guest experience extras from the brief's screenshots

The gallery has a search pill (where, check-in, check-out, guests), property-type chips, a wishlist heart and a photo carousel on every card. The property page shows the host with a Verified Host badge, rooms, amenities with icons, reviews and the booking card with a guest stepper.

5. Roles and access control

Area	Guest	Host	Admin	SuperAdmin
Browse, search and view properties	Yes	Yes	Yes	Yes
Book, check in, wishlist	Yes	No	No	No
Become a Host application	Yes	No	No	No
Host dashboard and bookings	No	Yes	No	No
Review host applications, bookings, properties, logs	No	No	Yes	Yes
Promote guests to Admin, reset passwords	No	No	No	Yes
Messages and notifications	Yes	Yes	Yes	Yes

A Host cannot open Guest-only pages and a Guest cannot open Host or Admin pages. Blocked users see an Access Denied page.

6. Data model

The model follows the entity diagram in the brief: ApplicationUser, HostProperty, Booking, Payment, Review, Amenity, PropertyImage, WishlistItem, Message, ActivityLog, Notification, HostApplication, CheckInProcess, GuestCheckIn and GuestDocument. Relationships: User to Properties, Bookings and Reviews (one to many); Property to Bookings, Reviews and Images (one to many); Booking to Payment and Booking to GuestCheckIn (one to one); Property to CheckInProcess (one to one); Property to Amenity (many to many); GuestCheckIn to GuestDocument (one to many).

Additions beyond the diagram: City, PropertyType, IsActive, CleaningFee, MaxGuests, Bedrooms, Beds and Bathrooms on HostProperty; NumberOfGuests on Booking; IsRead on Message and Notification.

7. Design decisions

The service fee is 12 percent of the nightly total and is set in one place (BookingPricing).

Payments are created as Pending with the booking request and become Paid when the host approves it. Host earnings and platform revenue count Paid payments only.

A user is either a Guest or a Host, never both. After a role change the user signs in again so the new role takes effect.

Guest ID documents are stored outside wwwroot (App_Data). Only the guest, that property's host and admins can open them.

Property photos are validated by type and size.

8. Known limitations

Guests cannot yet write reviews in the interface. The Review table and the host Reviews page are in place, and the demo data includes sample reviews.

The notification page has All and Unread filters only.

Payments are simulated: there is no real payment gateway.

Messaging is not real time (no SignalR), as allowed by the brief.

9. Credits

Demo photos: Unsplash (free to use under the Unsplash License). Photographers: Vojtech Bruzek, Joao Emanuel, Frames For Your Heart, VisualsOfDana, Marcel Strauss, Roberto Nickson and Maverick Frame.